INDIVIDUAL MAJOR PROJECT | MACHINE LEARNING INTERNSHIP

Machine Learning System for Emotion Recognition in Speech

Using MFCC Features and a CNN-BiLSTM-Attention Deep Learning Model

Submitted by	Ronia Rosario K
Task	Data collection, preprocessing, model implementation and F1-score evaluation
Dataset	RAVDESS (speech) - 1440 clips, 24 actors, 8 emotions
Final model	Experiment 3: Compact CNN-BiLSTM-Attention, checkpoint by validation Macro F1 (79,080 parameters)
Headline result	Test Macro F1 0.5307 | accuracy 56.67% (SVM baseline Macro F1 0.4464)
Framework	Python, librosa, TensorFlow / Keras, scikit-learn
Date	27 October 2026

Table of Contents

Table of Contents 2

Abstract 3

1. Introduction 4

1.1 Problem statement . . . 4

1.2 Objectives and where each is delivered . . . 4

1.3 Contributions . . . 4

2. Background on Speech Emotion Recognition 4

2.1 MFCC . . . 4

2.2 Delta and delta-delta features . . . 5

2.3 CNN, BiLSTM and attention . . . 5

2.4 Macro F1-score . . . 5

3. Dataset 5

4. Data Preprocessing 6

5. MFCC Feature Extraction 6

6. Proposed CNN-BiLSTM-Attention Architecture 8

7. SVM Baseline 8

8. Experimental Methodology 8

8.1 Actor-independent data split . . . 8

8.2 Training configuration (selected model) . . . 8

8.3 Evaluation metrics . . . 9

8.4 Experiment log . . . 9

9. Results and Comparison 10

9.1 Overall comparison . . . 10

9.2 Classification report (test set, selected model) . . . 10

9.3 Confusion matrix . . . 12

10. Discussion 12

10.1 What was observed . . . 12

10.2 Possible explanations (not experimentally tested) . . . 13

11. Limitations 13

12. Conclusion 13

13. Future Improvements 14

14. Reproducibility 14

Project structure . . . 14

References 14

Abstract

Speech carries emotional information in addition to words. This project builds and evaluates a machine learning system that recognises eight emotions (neutral, calm, happy, sad, angry, fearful, disgust, surprised) from short speech clips of the RAVDESS corpus. Each clip is resampled, trimmed, normalised and converted into **Mel-Frequency Cepstral Coefficients (MFCC)** with their first- and second-order derivatives (130 frames x 120 features). A compact **CNN-BiLSTM-Attention** network (79,080 parameters) learns local spectral patterns, temporal context and the most informative frames. The evaluation is **actor-independent**: actors 1-16 are used for training, 17-20 for validation and 21-24 for testing, so no voice appears in more than one partition.

On the held-out test actors the selected model (Experiment 3) obtains a **Macro F1 of 0.5307**, weighted F1 0.5384 and accuracy 56.67%, compared with a Macro F1 of 0.4464 for a tuned RBF-SVM baseline (+0.0843). Performance is uneven across emotions: recall is high for angry and surprised speech but low for sad and fearful speech. The results are obtained on acted speech from 24 actors and should not be read as a claim that the system reliably recognises emotion in everyday conversation.

Keywords: speech emotion recognition, MFCC, CNN, BiLSTM, attention, RAVDESS, Macro F1, actor-independent evaluation.

1. Introduction

Emotion-aware applications such as call-centre analytics, tutoring software, wellbeing assistants and human-robot interaction need to understand *how* something is said, not only *what* is said. Speech Emotion Recognition (SER) maps an audio signal to an emotion category. It is hard because emotion is subjective, every speaker expresses it differently, and public datasets are small, so models easily learn to recognise *voices* instead of *emotions*.

1.1 Problem statement

Develop a machine learning system for emotion recognition in speech using MFCC features and a deep learning model. The project should include data collection, preprocessing, model implementation, and evaluation using F1-score.

1.2 Objectives and where each is delivered

Objective	Where it is delivered
Data collection	ser/data.py - RAVDESS download, filename-based labelling, metadata table (Section 3)
Preprocessing	ser/features.py - resampling, silence trimming, normalisation, fixed 3 s window, augmentation (Section 4)
MFCC features	40 MFCC + delta + delta-delta = 120 features x 130 frames per clip (Section 5)
Deep learning model	ser/model.py - CNN + BiLSTM + temporal attention (Section 6)
Classical baseline	ser/train.py - RBF-SVM on utterance-level MFCC statistics (Section 7)
F1-score evaluation	ser/evaluate.py, ser/final.py - macro / weighted / per-class F1, confusion matrices (Sections 8-10)
Deliverables	Source code, tests, this report, presentation, README, results summary

1.3 Contributions

- A reproducible, tested pipeline with a fixed random seed, central configuration and automatic verification of every reported metric.
- An actor-independent split that prevents speaker leakage, a frequent flaw in SER studies.
- A compact CNN-BiLSTM-Attention model compared against a tuned classical baseline over several documented experiments.
- An honest error analysis that separates observed errors from plausible, untested explanations.

2. Background on Speech Emotion Recognition

2.1 MFCC

MFCCs (Davis and Mermelstein, 1980) describe the short-time spectral envelope of speech on the perceptual mel scale. The signal is cut into overlapping frames, each frame is converted to a power spectrum, passed through a mel filter bank, log-compressed and decorrelated with a discrete cosine transform. The low-order coefficients mainly describe the shape of the vocal tract, which changes with arousal and speaking style.

In plain words: MFCCs turn raw sound into a small table of numbers that describes the 'timbre' of the voice at every instant, similar to how the ear perceives it.

2.2 Delta and delta-delta features

The delta (first derivative) of each coefficient measures how fast it changes between frames; the delta-delta (second derivative) measures how that change accelerates. Emotion is expressed through dynamics - an angry voice changes abruptly, a sad voice slowly - so adding both derivatives gives the network information about movement, not only about static timbre.

2.3 CNN, BiLSTM and attention

A **1-D convolutional layer** slides small filters along time and detects short local patterns in the MFCC table. A **bidirectional LSTM** (Hochreiter and Schmidhuber, 1997) reads the sequence forwards and backwards and remembers context, for example a rising pitch followed by a pause. **Attention pooling** (Bahdanau et al., 2015, adapted here to a single utterance-level summary) lets the network learn which frames matter most and take a weighted average of them.

In plain words: The CNN spots short sound patterns, the BiLSTM understands how they follow each other, and attention decides which moments of the clip deserve the most weight in the final decision.

2.4 Macro F1-score

For each emotion, precision = TP / (TP + FP) (how often a predicted emotion is right), recall = TP / (TP + FN) (how many real examples are found) and F1 = 2PR / (P + R). **Macro F1** averages the F1 of all classes with equal weight, so a model cannot hide poor performance on one emotion behind good performance on others; this is why it is the primary metric here. **Weighted F1** weights classes by their number of test clips.

In plain words: Accuracy only counts how many answers are right overall. Macro F1 asks: is the system good at every emotion, not just the easy ones?

3. Dataset

The project uses the **Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS)** (Livingstone and Russo, 2018), available for research under CC BY-NC-SA 4.0. The speech subset has 1,440 recordings: 24 professional actors (12 female, 12 male) speak two matched sentences in eight emotions at two intensities (neutral has a single intensity, hence 96 neutral clips and 192 of every other emotion). Emotion, intensity, statement, repetition and actor are encoded in the filename, for example `03-01-05-02-01-01-07.wav` = speech, angry, strong intensity, actor 7. **Collection:** `python run_pipeline.py --download` downloads and unzips the archive; labels are parsed from filenames into a pandas metadata table.

Item	Value
Dataset	RAVDESS (speech)
Clips / actors / classes	1440 / 24 / 8
Training	960 clips, actors 1-16 (2880 samples after augmentation)
Validation	240 clips, actors 17-20 (not augmented)
Test	240 clips, actors 21-24 (not augmented)

Item	Value
Class counts	neutral 96, calm 192, happy 192, sad 192, angry 192, fearful 192, disgust 192, surprised 192
Random seed	42

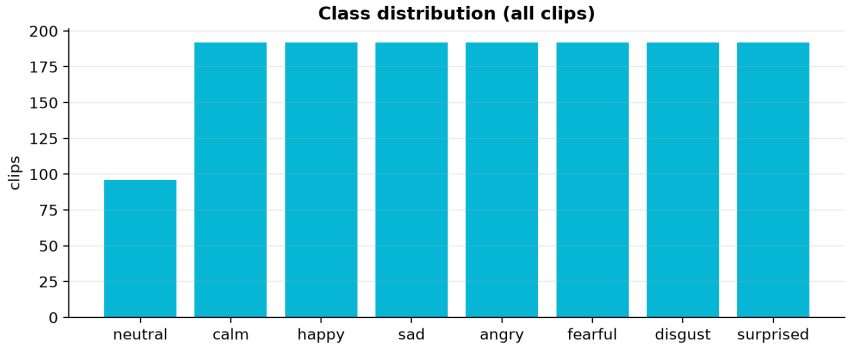


Figure 1 - Number of clips per emotion (neutral has half as many clips).

4. Data Preprocessing

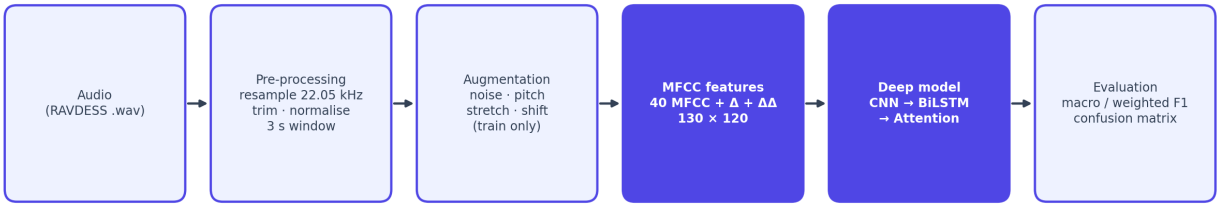


Figure 2 - End-to-end pipeline.

Every clip passes through the same steps:

- **Resampling** to 22,050 Hz mono (RAVDESS is recorded at 48 kHz).
- **Silence trimming** (30 dB threshold) and **peak normalisation** remove leading/trailing silence and loudness differences.
- **Fixed 3-second window**: centre crop or zero-padding, so every clip yields a tensor of the same shape.
- **Augmentation (training clips only)**: two extra copies per clip, each with one random transformation - Gaussian noise, pitch shift (+/-2 semitones), time stretch (0.9-1.1x) or time shift - giving 2,880 training samples. Validation and test clips are never augmented.
- **Standardisation**: per-feature mean and standard deviation are computed on the **training set only**, saved to `norm.npz`, and re-used unchanged for validation, test and inference.
- **Input validation**: unreadable, empty or silent audio raises a clear error instead of producing a silent wrong prediction.

5. MFCC Feature Extraction

Parameter	Value
Sampling rate	22050 Hz
Clip length	3.0 s
MFCC coefficients	40
FFT size / hop length	2048 / 512 samples
Mel bands	128
Features per frame	120 = 40 MFCC + 40 delta + 40 delta-delta
Tensor per clip	130 frames x 120 features

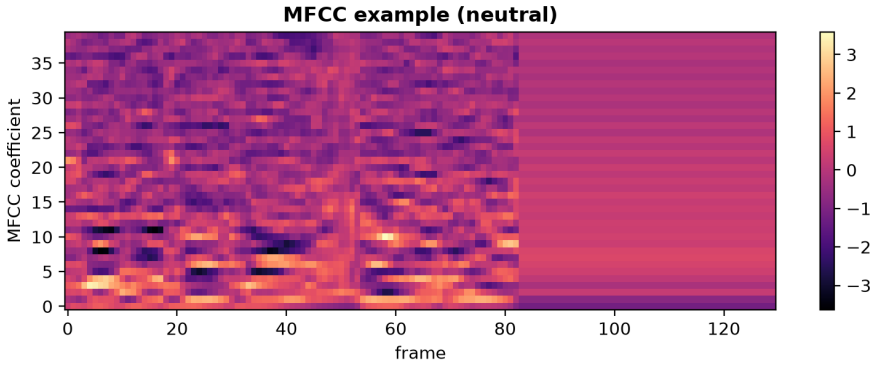


Figure 3 - MFCC matrix of a test clip (first 40 coefficients shown).

The tensor shape (130, 120) was confirmed from the saved experiment metadata and by the unit tests, which extract features from audio and check the shape and finiteness.

6. Proposed CNN-BiLSTM-Attention Architecture

The final model is the compact network of **Experiment 3** (79,080 trainable parameters). It processes the MFCC sequence in four stages:



Figure 4 - Architecture of the selected model (values read from the trained checkpoint).

- **Gaussian input noise** (sigma 0.08) acts as a regulariser during training only.
- **Two Conv1D blocks** (32 filters, kernel 5; 64 filters, kernel 3), each with batch normalisation, ReLU, max-pooling and dropout (0.3, 0.35); 130 frames are reduced to 32.
- **Bidirectional LSTM** (48 units per direction, dropout 0.3) returns one vector per time step.
- **Additive attention pooling** (32 units): a small tanh network scores every frame, a softmax turns the scores into weights, and the weighted sum is the utterance embedding.
- **Classifier**: Dense(64, ReLU, L2 0.0002) - Dropout 0.5 - Dense(8, softmax) giving eight probabilities that sum to one.

Loss is sparse categorical cross-entropy with balanced class weights; the optimiser is Adam. The attention layer is a registered custom Keras layer so that checkpoints save and load reliably (a Keras notice about its un-built sub-layers appears when loading; the model nevertheless loads, predicts and reproduces the training-time outputs).

7. SVM Baseline

A classical baseline shows how much the deep model adds. Every clip is summarised by the mean and standard deviation of each of the 120 features over time (240 numbers). A standardiser (fitted on training data only) and an RBF-kernel SVM with balanced class weights are trained on the same actor-independent training set. The hyper-parameters C in {1, 10, 100} and gamma in {scale, 0.001, 0.01} are chosen by Macro F1 on the **validation actors only** (selected: C = 100, gamma = 0.001); the final baseline is then evaluated once on the test actors. It uses the same class-label order as the deep model.

8. Experimental Methodology

8.1 Actor-independent data split

Actors 1-16 form the training set, 17-20 the validation set and 21-24 the test set. The code asserts that no actor appears in two partitions. A random clip-level split would put the same voices in training and test data and overstate performance, because the model could recognise the speaker instead of the emotion.

8.2 Training configuration (selected model)

Setting	Value
Optimiser / learning rate	Adam, 1e-3; halved after 4 epochs without validation-loss improvement (minimum 1e-5)
Loss / class weights	Sparse categorical cross-entropy / balanced
Batch size / maximum epochs	32 / 60
Checkpoint selection	validation Macro F1 (best epoch 12 of 24 run)
Early stopping	12 epochs without validation Macro F1 improvement
Random seed	42
Training time	170 s on CPU
Software (project environment)	Python 3.13, TensorFlow 2.21.0, Keras 3.15.1, librosa 1.0.0, scikit-learn

8.3 Evaluation metrics

On the test actors the report gives accuracy, macro precision, macro recall, Macro F1, weighted F1, per-class precision/recall/F1 and the confusion matrix, always over all eight labels. Every figure is recomputed from the saved confusion matrix and checked against the stored value by `summarize_results.py` and by the unit tests.

8.4 Experiment log

Experiment	Change	Params	Val Macro F1	Test Macro F1
Experiment 1	First full model (3 Conv1D blocks, BiLSTM 96, Dense 128); checkpoint by validation loss.	341,064	0.4484	0.4292
Experiment 2	Compact network (2 Conv1D blocks, BiLSTM 48, Dense 64), stronger regularisation; checkpoint by validation loss.	79,080	0.4873	0.5231
Experiment 3	Same compact network; checkpoint and early stopping on validation Macro F1; up to 60 epochs.	**79,080**	**0.5161**	**0.5307**
Experiment 4	Compact network with slightly stronger dropout (0.35 / 0.40 / 0.35 / 0.50).	79,080	0.4794	0.4783

Experiment 4 (completed, single run, seed 42): validation Macro F1 0.4794, test Macro F1 0.4783 versus 0.5161 / 0.5307 for Experiment 3. Stronger dropout did not improve generalisation in this run, so Experiment 3 remains the final model.

9. Results and Comparison

9.1 Overall comparison

Model	Val Macro F1	Test acc.	Test Macro F1	Test weighted F1
Experiment 1 - deep model	0.4484	0.4708	0.4292	0.4323
Experiment 2 - deep model	0.4873	0.5375	0.5231	0.5343
Experiment 3 - deep model	**0.5161**	**0.5667**	**0.5307**	**0.5384**
Experiment 4 - deep model	0.4794	0.5000	0.4783	0.4769
SVM baseline (MFCC statistics)	-	0.4667	0.4464	0.4624

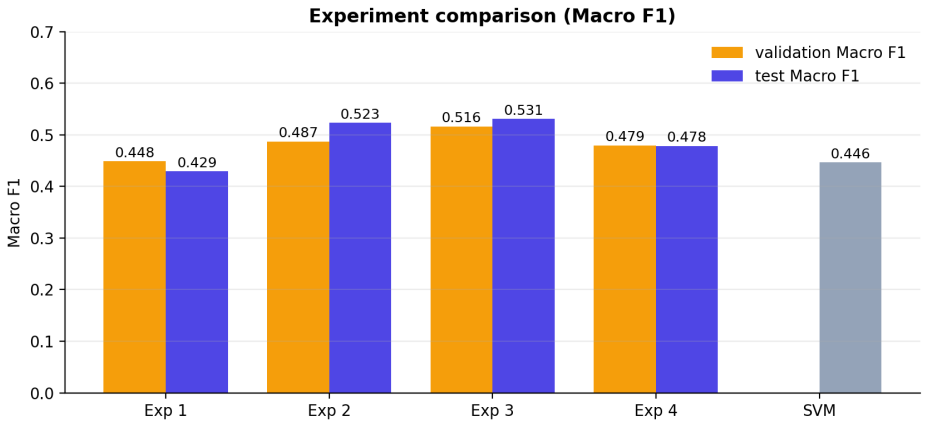


Figure 5 - Validation and test Macro F1 of every completed experiment and the SVM baseline.

Selected final model: Experiment 3. It has the highest validation Macro F1 (0.5161) of the completed experiments, so it ranks first under the validation-only selection rule used for this project; it also has the highest test Macro F1. Its test Macro F1 is +0.0843 higher than the SVM baseline (macro precision 0.5743, macro recall 0.5742). Because the test set has only 240 clips from 4 actors, differences of a few F1 points between experiments are within plausible sampling noise and should not be over-interpreted. Note also that Experiments 1-3 were compared while test results were visible, so the test figure is a slightly optimistic estimate; the validation ranking points in the same direction.

9.2 Classification report (test set, selected model)

Emotion	Precision	Recall	F1	Support
neutral	0.2973	0.6875	0.4151	16
calm	0.5897	0.7188	0.6479	32
happy	0.4783	0.3438	0.4000	32
sad	0.4545	0.1562	0.2326	32
angry	0.5918	0.9062	0.7160	32
fearful	0.8000	0.2500	0.3810	32
disgust	0.6111	0.6875	0.6471	32
surprised	0.7714	0.8438	0.8060	32
macro average	**0.5743**	**0.5742**	**0.5307**	**240**

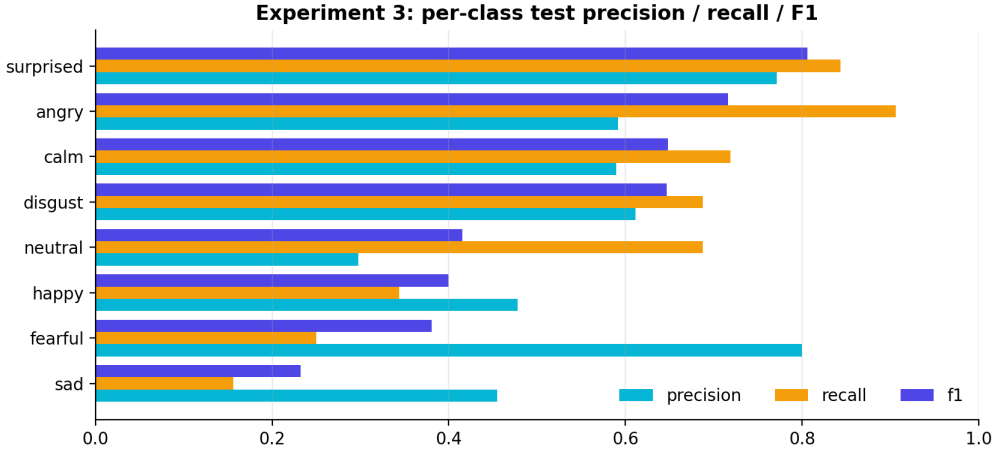


Figure 6 - Per-class precision, recall and F1 on the test actors.

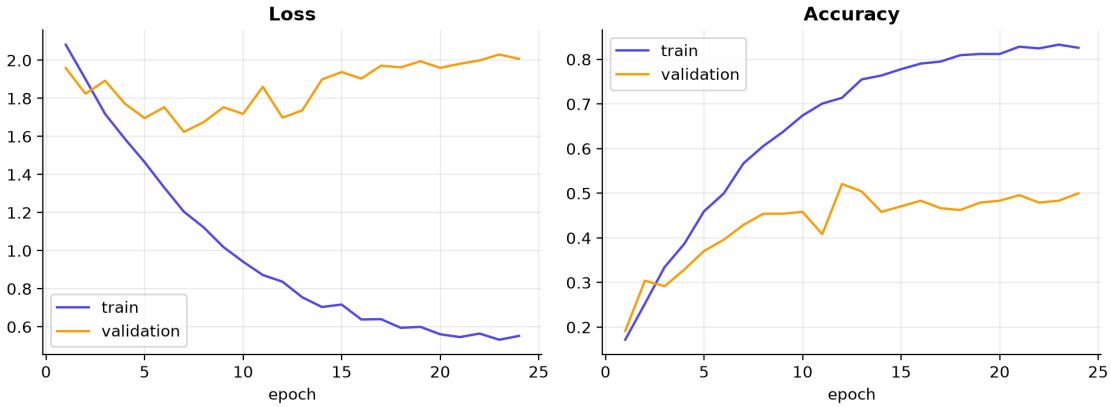


Figure 7 - Training and validation loss / accuracy of the selected run.

9.3 Confusion matrix

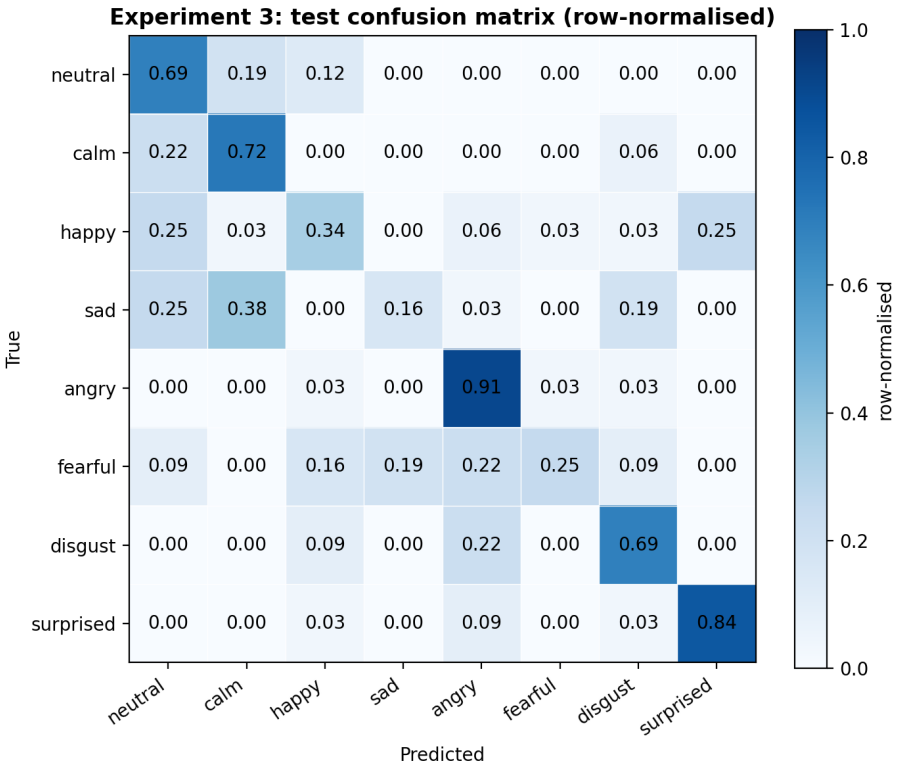


Figure 8 - Row-normalised confusion matrix of the selected model (rows: true emotion, columns: predicted).

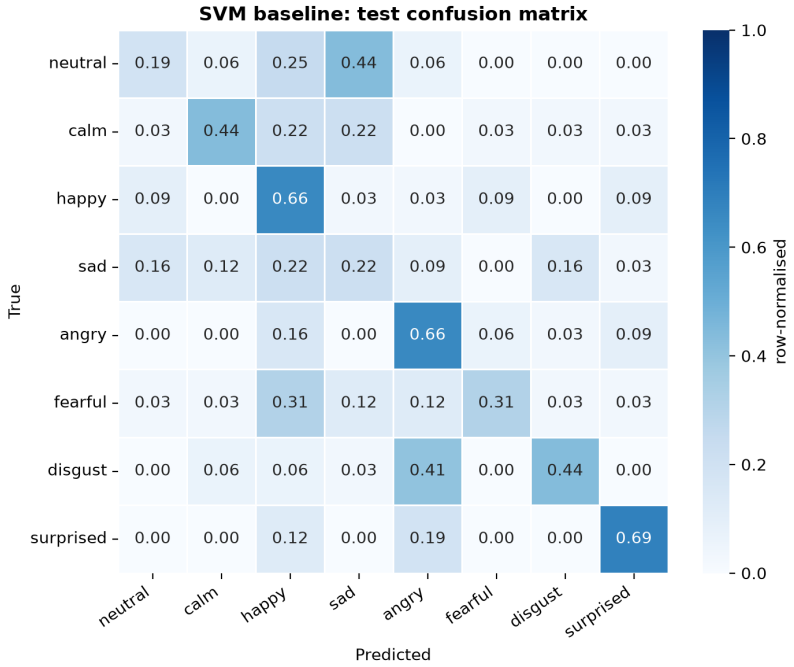


Figure 9 - Confusion matrix of the SVM baseline for comparison.

10. Discussion

10.1 What was observed

- The best recognised emotion was **surprised** (F1 0.8060); the weakest was **sad** (F1 0.2326, recall 0.1562).
- Lowest recalls: **sad** 0.1562, **fearful** 0.2500, **happy** 0.3438. Highest recall: **angry** 0.9062.
- **Neutral is over-predicted**: the model assigned 37 test clips to neutral although only 16 are neutral, giving neutral a precision of just 0.2973.
- Most frequent confusions (true -> predicted): sad -> calm (12 clips, 38%); happy -> neutral (8 clips, 25%); happy -> surprised (8 clips, 25%); sad -> neutral (8 clips, 25%); calm -> neutral (7 clips, 22%); fearful -> angry (7 clips, 22%).
- **Fearful** speech is rarely predicted wrongly (precision 0.8000) but is often missed (recall 0.2500): the model is cautious about this class.

10.2 Possible explanations (not experimentally tested)

The following are plausible hypotheses, not proven causes. (i) Sad, calm and neutral speech share low energy and slow, flat delivery, so MFCC-based features may overlap; this would fit the confusions of sad with calm and neutral. (ii) Angry, disgust, fearful and happy speech can share high arousal, which may explain confusions between them. (iii) With only 16 training actors the model may learn actor-specific habits, which would hurt unseen actors. (iv) Four test actors give a noisy estimate, so a few clips shift a class recall noticeably. Testing these hypotheses would require additional experiments (for example per-actor analysis or cross-validation over actors).

11. Limitations

- **Small, acted dataset**: 1,440 clips, 24 actors, two sentences; natural, spontaneous or noisy speech was not evaluated, and the model must not be assumed to understand human emotion in general.
- **Unseen speakers**: generalisation is measured on only four test actors; an unlucky or lucky actor split can move the scores.
- **Class imbalance**: neutral has half as many clips as the other classes (handled with class weights, which does not remove the effect completely).
- **Test-set exposure**: Experiments 1-3 were compared with test results visible, so the final test score is mildly optimistic (Experiment 4 was selected or rejected by the validation-only rule).
- **Features**: only MFCC-based features were used, as required; no pitch/energy features, no noise-robustness or cross-corpus tests.
- **Experiment 4 is a single run**: the stronger-dropout variant was trained once with one seed, so its difference from Experiment 3 may be within run-to-run variation.
- **Ethics**: emotion recognition can be misused for surveillance or high-stakes decisions; this system is a research prototype and no demographic fairness analysis was carried out.

12. Conclusion

The project delivers every element of the problem statement: data collection from RAVDESS, a documented preprocessing and MFCC pipeline, a CNN-BiLSTM-Attention implementation and an F1-centred evaluation under an actor-independent protocol. The selected model reaches a test Macro F1 of 0.5307 (accuracy 56.67%), clearly above the SVM baseline (0.4464) and the first deep model, but performance is far from perfect and uneven across emotions, particularly for sad and fearful speech.

The results are an honest, reproducible baseline for acted speech rather than evidence of general emotion understanding.

13. Future Improvements

- Repeat the Experiment 3 / 4 comparison with several random seeds to separate real gains from run-to-run variation.
- Actor-grouped k-fold cross-validation with confidence intervals for a more reliable estimate.
- SpecAugment-style masking, mixup, and additional corpora (TESS, CREMA-D, EMO-DB) for cross-corpus evaluation.
- Additional prosodic features (pitch, energy) or pre-trained speech encoders, evaluated with the same protocol.
- A real-time microphone demo and an API, with explicit user-consent and privacy safeguards.

14. Reproducibility

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install -r requirements.txt
python -m pytest -q
python summarize_results.py
python predict.py path\to\clip.wav
python build_report.py --name "Your Name"
python build_ppt.py --name "Your Name"
```

Project structure

ser/config.py - settings | ser/data.py - download, labels, split | ser/features.py - preprocessing, MFCC, augmentation | ser/model.py - Experiment 1-3 network | ser/model_exp4.py - Experiment 4 network | ser/train.py, ser/train_exp4.py - training | ser/evaluate.py - metrics and plots | ser/final.py - result verification and model selection | predict.py - inference | tests/ - automated tests

References

[1] Livingstone, S. R. and Russo, F. A. (2018). The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS). PLoS ONE 13(5): e0196391.

[2] Davis, S. and Mermelstein, P. (1980). Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences. IEEE Trans. ASSP 28(4), 357-366.

[3] Hochreiter, S. and Schmidhuber, J. (1997). Long short-term memory. Neural Computation 9(8), 1735-1780.

[4] Kingma, D. P. and Ba, J. (2015). Adam: A method for stochastic optimization. ICLR.

[5] Ioffe, S. and Szegedy, C. (2015). Batch normalization: accelerating deep network training by reducing internal covariate shift. ICML.

[6] Srivastava, N. et al. (2014). Dropout: a simple way to prevent neural networks from overfitting. JMLR 15, 1929-1958.

[7] Bahdanau, D., Cho, K. and Bengio, Y. (2015). Neural machine translation by jointly learning to align and translate. ICLR.

[8] McFee, B. et al. (2015). librosa: Audio and music signal analysis in Python. Proc. SciPy.

[9] Pedregosa, F. et al. (2011). Scikit-learn: machine learning in Python. JMLR 12, 2825-2830.

[10] Akcay, M. B. and Oguz, K. (2020). Speech emotion recognition: emotional models, databases, features, preprocessing methods, supporting modalities, and classifiers. Speech Communication 116, 56-76.